Requirements Gathering

Project: LinkedIn Job Market Analysis — What Drives Demand and Pay for Data Roles
Track: Junior Data Analysis — DEPI **Version / Date:** 1.0 — 15 Oct 2026

1. Introduction

This document records what the system (a Power BI dashboard on LinkedIn job-posting data) must do and the data it relies on. Requirements were gathered through **dataset profiling** (structure, coverage and quality of all 11 files), **review of similar dashboards and Kaggle analyses**, and **persona-based user stories** for the people expected to use the dashboard.

2. Stakeholders and Personas

Stakeholder	Need
Job seeker / fresh graduate	See which roles, skills and industries are in demand and what they pay
Career advisor / trainer	Evidence to guide students on skills to learn
Recruiter / HR analyst	Benchmark salaries, benefits and competition for roles
Project team and DEPI evaluators	A correct, documented, reproducible analysis

3. Business Requirements

- **BR-1:** Understand which job titles, industries and locations have the highest demand.
- **BR-2:** Understand what drives salary (experience level, industry, skills, work type, remote option).
- **BR-3:** Identify the most requested skills and benefits.
- **BR-4:** Measure competition per job (views, applications, apply rate).
- **BR-5:** Provide a focused view of Data / Business Analyst roles.

4. Functional Requirements

ID	Requirement	Priority
FR-1	Load and combine the 11 CSV files into one data model	Must
FR-2	Clean data: remove duplicates, convert timestamps, standardise pay to a yearly figure, flag salary outliers	Must
FR-3	Overview page with KPI cards: total jobs, total companies, average and median salary, remote %	Must
FR-4	Salary page: salary by experience level, industry, state and work type; salary distribution	Must
FR-5	Skills & Benefits page: top skills, skills vs salary, most common benefits	Must
FR-6	Industries & Companies page: top industries and employers, company size vs hiring	Must
FR-7	Demand & Competition page: views, applications, apply rate, sponsored vs non-sponsored	Must
FR-8	Data Analyst deep-dive page	Must
FR-9	Slicers for work type, experience level, industry, state and date	Must
FR-10	Drill-through from industry to company and job title	Should
FR-11	Tooltips with salary coverage (% of jobs with salary) on salary visuals	Should
FR-12	Export of summary insights as a report	Could

5. Non-Functional Requirements

ID	Requirement
NFR-1	**Performance:** pages respond to filters in under 5 seconds

ID	Requirement
NFR-2	**Accuracy:** totals reconcile with the source data (jobs counted with DISTINCTCOUNT on job_id)
NFR-3	**Usability:** consistent theme, clear titles, navigation buttons between pages
NFR-4	**Reproducibility:** cleaning scripts and steps documented on GitHub
NFR-5	**Transparency:** every salary visual states its coverage and filters used
NFR-6	**Privacy:** only public, non-personal data is used

6. Data Requirements

Table	Rows	Key fields
postings	123,849	job_id, title, company_id, work type, experience level, remote_allowed, normalized_salary, views, applies, listed_time
salaries	40,785	job_id, min / med / max salary, pay_period, currency
job_skills	213,768	job_id, skill_abr
job_industries	164,808	job_id, industry_id
benefits	67,943	job_id, benefit type
companies	24,473	company_id, size, state, country, city
company_industries	24,375	company_id, industry
company_specialities	169,387	company_id, speciality
employee_counts	35,787	company_id, employee and follower counts
industries / skills (lookups)	422 / 35	ids and names

Data period: 24 Mar 2024 – 20 Apr 2024 (about one month).

7. Data Quality Findings (from profiling)

Issue	Finding	Handling
Salary coverage	Only about 29% of postings have a normalised salary	Analyse the salary subset only and display coverage
Salary outliers	Extreme values (maximum in the hundreds of millions)	Filter to a realistic yearly range and document the rule
Remote flag	About 88% missing	Do not draw conclusions on remote from this field alone
Applications	About 81% missing	Calculate apply rate only where available
Timestamps	Stored as milliseconds	Convert to date
Many-to-many links	Jobs have several skills and industries	Use bridge tables and distinct counts
Large file	postings is about 500 MB with multi-line descriptions	Keep only needed columns; exclude description text
Geography	Mostly US locations	State this limit; no Egypt-specific conclusions

8. User Stories

- As a **job seeker**, I want to filter by experience level so that I can see entry-level roles and their pay.
- As a **job seeker**, I want to see the top skills for Data Analyst roles so that I know what to learn.
- As a **career advisor**, I want to compare skills by salary so that I can recommend high-value skills.
- As a **recruiter**, I want to see typical benefits and salary by industry so that I can benchmark offers.
- As an **evaluator**, I want totals that match the source data so that I can trust the dashboard.

9. Assumptions and Constraints

- The dataset is a public Kaggle snapshot and is treated as representative of the period only.
- The project uses Power BI Desktop and Python; no paid services.
- The deadline schedule follows the DEPI documentation timetable.
- Descriptions are not mined in the base scope.

10. Acceptance Criteria

Requirement	Accepted when
FR-1, FR-2	All 10 tables load; no duplicate job_id in the fact table; outlier rule documented
FR-3 to FR-8	Six pages built with the visuals listed above
FR-9	Slicers filter every relevant visual on each page
NFR-1	Filters respond in under 5 seconds on the full data
NFR-2	Dashboard job count equals the source count (123,849)
BR-1 to BR-5	At least 7 insights documented with evidence from the dashboard